Alieyeh Sarabandi Moghaddam

Health Data Researcher | Longitudinal Mental Health Data | Study Coordination

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Profile

Health data researcher with an MSc in Health Data Science and four years of experience across dementia research, clinical machine learning, human genetics and secure research data management. Experienced in longitudinal and multimodal data cleaning, database maintenance, statistical and machine-learning analysis, research documentation, publication and coordination across clinical, academic, governance and technical teams.

Current work at Dementias Platform UK (DPUK) combines sensitive cohort-data management with independent dementia research. I conceived and developed a multi-omic dementia subtyping project with guidance and collaboration from Dr Esteban Gomez and Professor Alfredo Iacoangeli at King's College London. I bring careful organisation, clear communication, public-engagement experience and a strong commitment to respectful, participant-centred research.

Relevant Skills

Longitudinal and remote-monitoring data: clinical and cohort data, repeated measures, HAR, EEG and ECG time series, missingness, harmonisation, metadata alignment, quality control and validation

Data cleaning and analysis: Python, R, SQL, pandas, scipy, scikit-learn, PyTorch, descriptive analysis, feature engineering, classification, clustering, model evaluation and visualisation

Databases and research data management: PostgreSQL, SQL Server, SQLite, MongoDB, SQL querying, relational schemas, data dictionaries, validation, reconciliation, provenance, audit trails and dashboard maintenance

Research delivery and organisation: concurrent workstream management, study documentation, SOPs, task tracking, meeting contributions, issue escalation, reporting, deadlines and intern supervision

Communication and engagement: peer-reviewed writing, protocols, conference presentations, researcher guidance, an HPC training course and public-facing workshops on AI safety and sensitive health data

Mental health and biomedical research: dementia heterogeneity, psychiatric and neurological PRS, clinical outcomes, longitudinal cohorts, multi-omics and responsible interpretation

Sensitive data and governance: Trusted Research Environments, confidentiality, GDPR-aware research, disclosure control, ethical approvals, PII checking and ONS Safe Researcher accreditation

Research and Professional Experience

Genomic Data Scientist — **2023 – Present**
Dementias Platform UK (DPUK), Swansea University — *Swansea, UK*

- Support research across dementia cohorts by preparing genomic, multi-omic, longitudinal and clinical-context data for reliable use within a Trusted Research Environment.
- Clean, validate and harmonise heterogeneous datasets; investigate missing or inconsistent records, check identifiers and metadata, maintain provenance and document limitations for researchers.
- Design and maintain relational schemas, metadata models, data dictionaries, tagging systems and discovery structures; develop and validate SQL for PostgreSQL- and SQL Server-based research systems.
- Maintain the Great Minds Feasibility Tool, troubleshooting database scripts and scheduled updates that feed a Power BI dashboard used to communicate cohort and data availability to researchers.
- Built Python- and SQLite-based integrity-monitoring workflows that maintain transactional records and report unexpected deletion, modification, movement or version changes across secure research storage.
- Coordinate dataset onboarding, secure provisioning, project queries, output checks and documentation across researchers, data managers, data owners, governance specialists and infrastructure teams.
- Manage multiple concurrent priorities using Jira, Confluence and structured technical records; raise risks early, track dependencies and communicate progress clearly.
- Produce SOPs, researcher guidance, reports, manuscripts and conference materials; developed a practical course on secure HPC use, workflow execution, output checking and responsible data handling.
- Supervise an intern working on omics metadata discovery infrastructure, including task planning, feedback, review and support for independent problem-solving.
- Contribute to privacy-preserving AI and NLP work involving PII detection, synthetic data, model evaluation and responsible use of sensitive participant information.

Data Scientist – Human Genetics **2023**
GlaxoSmithKline (GSK), MSc Industry Placement *UK*

- Conducted scientific research using UK Biobank whole-genome sequencing data in a regulated pharmaceutical environment.
- Developed Python workflows for data import, annotation, cleaning, aggregation and validation of rare non-coding genomic regions and performed region-based association analysis.
- Managed an independent MSc research project, maintained reproducible records and communicated methods and results to an international team of geneticists, statisticians and bioinformatics scientists.

Health Data Scientist – Clinical Machine Learning Research **2021 – 2022**
Professor Maryam Khademi's Research Group *Tehran, Iran*

- Analysed structured hospital data collected during the COVID-19 outbreak to investigate mortality among patients with diabetes.
- Completed data cleaning, feature engineering, statistical and machine-learning analysis, model evaluation and interpretation under ethical approval.
- Co-authored the resulting peer-reviewed publication in the *Journal of Diabetes & Metabolic Disorders*.

Selected Research Projects

Precision Subtyping of Dementia Through Multi-Omic Integration *Independent research idea with KCL collaboration*

- Conceived the project and developed its analytical framework to investigate dementia heterogeneity using genomic, multi-omic, longitudinal and clinical cohort data.
- Develop the work with guidance and collaboration from Dr Esteban Gomez and Professor Alfredo Iacoangeli at King's College London, alongside the wider DPUK research context.
- Methods include cross-cohort harmonisation, missingness and batch-effect assessment, feature engineering, clustering, pathway analysis and validation against clinical endpoints.
- First-author poster presented at the Alzheimer's Research UK Conference in 2026; study protocol manuscript in preparation.

Multimodal Remote-Monitoring Data Pipeline *Independent open-source project*

- Built a reproducible Python pipeline for cleaning, harmonising and validating longitudinal HAR, EEG and ECG datasets, including variable sampling structures, time-series segmentation, aligned metadata and fixed-shape outputs.
- Added manifests, validation reports, resource estimates, resumable execution and automated tests to support reliable downstream analysis.

HYPEHD Longitudinal Health Data Package *PyPI and GitHub*

- Developed and released a reusable Python package for exploring longitudinal and demographic health data, with documented analysis and visualisation utilities.

Education

MSc Health Data Science, Merit 2022 – 2023
University of Exeter
GSK-linked thesis: *Using human genetic data to inform drug discovery through region-based association testing*

BSc Computer Engineering, equivalent to UK 2:1 2016 – 2021
Iran University of Science and Technology
Thesis: *Disease prediction using symptom–disease network modelling and probabilistic inference software*

Selected Publications and Research Communication

- First author of *Choosing the Right Genomic Dataset*, *Real World Data Science*, 2026.
- Co-author of a peer-reviewed clinical machine-learning study in the *Journal of Diabetes & Metabolic Disorders*, 2023.
- First-author dementia subtyping poster, Alzheimer's Research UK Conference, 2026.
- Accepted first-author presentation on genomics data infrastructure and cohort discovery, IPDLN Conference, 2026.
- Second-author accepted poster on an NLP tool for automated detection of participant identifiers, IPDLN Conference, 2026.
- Contributor to DARE UK reports and public-facing workshops on safe AI, synthetic data, privacy and responsible health-data research.